B.Tech. Degree IV Semester Examination, May 2006

ME 403 APPLIED THERMODYNAMICS

(2002 Admissions)

Time: 3 Hours | Maximum Marks: 100

I a) Explain the following ***terms***:
   i) Enthalpy   ii) Internal energy
   iii) Reversibility (3)

b) Prove that energy is a property. (5)

c) In a steam power plant the boiler is supplied with 60 kg of water per minutes. The enthalpy and velocity of water entering the boiler are 840 KJ/kg and 5m/s respectively. The water received 2300 KJ/kg of heat at constant pressure in the boiler. The boiler inlet is 3m above the turbine exit. The steam leaves the turbine with a velocity of 3000m/min and the enthalpy is 2640KJ/kg. The heat losses from the boiler and turbine to the surroundings is 1260 KJ/min. Determine the power output of the turbine. (12)

OR

II a) State the Kelvin Planck and Clausius statements of the second law of thermodynamics and prove their equivalence. (8)

b) The air-conditioning plant of a cinema house having a seating capacity of 1200, is run by a carnot refrigerator. The temperature in the cinema house is maintained at 25°C and the atmospheric temperature is 38°C. The carnot refrigerator is run by a carnot engine which absorbs heat from a source at a temperature of 107°C and rejects it to the atmosphere. Determine the required capacity of the carnot engine in KW and the amount of fuel of calorific value 22000 KJ/kg required to be burned per hour for running the engine. The amount of heat dissipated per person is 418 KJ/hour. (12)

III a) State Dalton's law of partial pressures. (10)

b) The specific heats of a gas are given by $C_p=a+kT$ and $C_v=b+kT$ where a, b and k are constants and T is temperature in K. Show that for an isentropic expansion of this gas $T^b v^{a-b} e^{kT}$= Constant. (10)

OR

IV a) Derive an expression for Gibb's function of a mixture of inert ideal gases at temperature T and pressure p. (10)

b) A mixture of ideal gases consists of 2 kg of nitrogen and 4 kg of carbon dioxide at a pressure of 300 kPa. Find:
   i) the mole fraction of each constituent
   ii) the equivalent molecular weight of the mixture
   iii) the equivalent gas constant of the mixture (10)

V a) Give a note on the various types of solid, liquid and gaseous fuels used in engineering practice. (10)

b) A sample of coal contains 78% Carbon, 6% Hydrogen, 1.2% Nitrogen, 7.8% Oxygen, 3% Sulphur and 4% incombustibles. Find:
   i) minimum quantity of air required for complete combustion of 1kg of coal
   ii) the total mass of products of combustion. (10)

OR

VI a) Describe the Orsat apparatus used for the gas analysis. (10)

b) Explain how you determine the percentage of carbon in fuel burning to CO and CO_2 from the volumetric composition of dry flue gas. (10)

(Turn Over)

VII	a)	Describe the working of four stroke and two stroke IC engines.	(10)
	b)	Explain how and why the Morse test is conducted on a multi cylinder S.I engine.	(10)

OR

VIII	a)	Explain the term brake mean effective pressure.	(5)
	b)	The following observations were taken during a test on a single cylinder four stroke cycle engine.	

Bore	-	30cm
Stroke	-	45 cm
Speed	-	300 rpm
Total fuel consumption	-	11.4 kg/hour
Calorific value of fuel	-	42000 kJ/kg
Indicated mean effective pressure	-	6 bar
Net Brake load	-	1.5 kN
Brake drum diameter	-	1.8 m
Brake rope diameter	-	2 cm

Determine (i) indicated power (ii) brake power (iii) mechanical efficiency (iv) indicated thermal efficiency. (15)

IX	a)	Explain how detonation can be controlled in SI engines.	(10)
	b)	Discuss combustion chamber design in SI engines.	(10)

OR

X	a)	What is diesel knock? How can it be prevented?	(8)
	b)	Write ***short notes*** on:	

i) Super changing of SE CI engines
ii) free piston engines
iii) cooling system in IC engines
iv) lubrication system in engines (12)

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